Five Small Errors in *Bayesian Persuasion*

The Theory Holds. Some of the Fine Print Does Not

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Abstract

Kamenica and Gentzkow (2011, *AER*) is the founding paper of information design. We checked its results line by line. The main theorems are correct: with finitely many states, the sender's best signal is found by concavifying his payoff, an optimal signal exists, and it needs no more signal realizations than there are states. Five statements are wrong as written. (1) The web appendix's extension to infinitely many states uses the wrong hull: in an example that meets all its assumptions, the formula gives the sender 0 while full revelation gives him 1. (2) A rule of thumb in the main text says that if the receiver's action and the sender's payoff are both concave, persuasion cannot help; it can. (3) In the lobbying application, "no disclosure is uniquely optimal" is false; informative studies do equally well. (4) A worked example uses the wrong formula. (5) Proposition 2's genericity claim relies on a stronger assumption than the one the paper states. Each has a one-line fix.

1 Setup and what survives

A sender chooses what a receiver will learn about a state ω. The receiver forms a belief μ and acts. Let $\hat{v}(\mu)$ be the sender's expected payoff when the receiver holds belief μ. The paper's central result is that the sender's best value is the concave closure of $\hat{v}$ at the prior: the lowest concave function lying above $\hat{v}$. With finitely many states this is correct, the optimum is attained, and at most as many signal realizations as states are needed. An exact-arithmetic check on 200 randomly drawn finite games agrees. Nothing below touches these results.

2 The five errata

1. Infinitely many states: the hull needs a closure. Section 3 of the web appendix extends the result to compact metric state spaces. It defines the value as the height of the *convex hull* of the hypograph of $\hat{v}$ above the prior, explicitly the intersection of all convex sets containing it, with no closure. Such a hull only contains *finite* mixtures of beliefs. The proof then asserts that this set is compact and that no signal can do better. Both claims fail.

Example (all of the appendix's assumptions hold). The state is uniform on $[0, 1]$; an extra isolated state, 2, has prior probability zero. The receiver either plays safe (payoff 0) or guesses a number $a \in [0, 1]$, with payoff $-(a - \omega)^2$, or 1 if $\omega = 2$. The sender gets 1 whenever the receiver guesses. Any finite signal leaves the receiver uncertain, so a guess has negative expected payoff and she plays safe: the appendix formula gives the sender 0. Revealing the state exactly makes her indifferent, she guesses (ties go to the sender, as the paper assumes), and the sender gets 1. Finer partitions do not help: a partition into n cells still leaves variance $1/(12n^2) > 0$ (Figure 1, right).

The fix: use the *closed* convex hull, so that the value is the smallest upper semicontinuous concave function above $\hat{v}$. The appendix's existence claim survives with this fix. The example is knife-edge, since it needs exact indifference, but the stated formula is wrong. It matters because later work cites it, for example Dworczak and Kolotilin's duality paper.

2. Concave plus concave is not concave. After Remark 1 the paper says that, when the sender cares only about the action, "if both [the receiver's action as a function of beliefs] and [the sender's payoff as a function of the action] are concave, Sender does not benefit from persuasion," and that if both are convex, he does. Composing concave functions preserves concavity only if the outer function is increasing.

Counterexample. Two states, $p = \Pr(\omega = 1)$, action $a \in [0, 1]$. The receiver's payoff $3a\omega - a^3$ gives action $\sqrt{p}$ (concave). The sender's payoff is $-a$ (concave). His payoff in beliefs is $-\sqrt{p}$, which is convex. At $p = 1/4$ silence gives $-1/2$ and full disclosure $-1/4$ (Figure 1, left). The convex case fails in mirror image: with receiver payoff $2\omega\sqrt{a} - a$ the action is p^2 (convex), the sender's payoff $-a$ is linear, and $-p^2$ is concave, so persuasion cannot help. Fix: add "and the sender's payoff is increasing in the action."

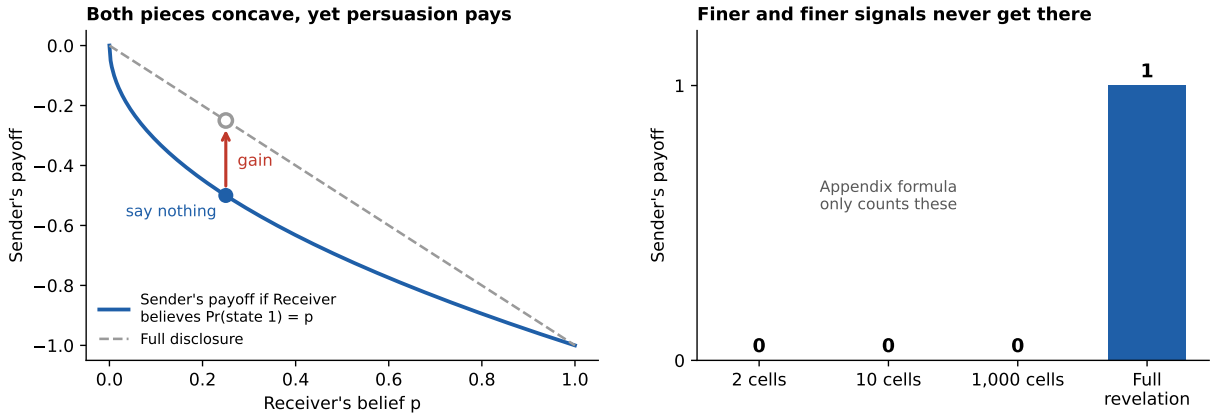


Figure 1: Left: erratum 2. The receiver's action $\sqrt{p}$ and the sender's payoff $-a$ are both concave, but the sender's payoff as a function of the receiver's belief, $-\sqrt{p}$, is convex, so full disclosure beats silence. Right: erratum 1. Sender's payoff in the web-appendix example from partitioning $[0, 1]$ into equal cells, and from full revelation.

3. Lobbying: silence is optimal, but not uniquely. The paper shows the lobbyist's payoff equals a constant plus $(2\alpha - 1)(E_\mu[\omega])^2$ and concludes that for $\alpha < 1/2$ "no disclosure is uniquely optimal." This payoff depends on the belief only through its mean, so it is concave but not strictly concave in beliefs. Any study that never moves the politician's expected state does exactly as well. With a prior symmetric around $1/2$, for example, a study that only reveals whether ω lies near the middle or near the ends is informative and optimal. Fix: for $\alpha < 1/2$ a signal is optimal if and only if every posterior mean equals the prior mean. The full-disclosure result for $\alpha > 1/2$ is unaffected.

4. A worked example ignores the action constraint. To show that a bound is not tight, the paper takes states $\{-1, 0, 1\}$, actions $a \in [0, 1]$, receiver payoff $-(a - \omega)^2$ and sender payoff a^2. It writes the sender's payoff as $(E[\omega])^2$, whose concave closure is constant at 1. Because actions cannot be negative, the receiver picks $\max(0, E[\omega])$, so the payoff is $\max(0, E[\omega])^2$, and the bound is $(1 + E[\omega])/2$, not 1. With a uniform prior the bound is $1/2$ and the true optimum is $1/3$. The example's point, that the bound can be strict, survives.

5. Proposition 2 uses an assumption the paper does not make. The paper assumes every action is optimal at *some* belief. The proof that the receiver is generically not indifferent uses the stronger assumption that every action is *uniquely* optimal at some belief. Under the weaker one, two actions the receiver values identically but the sender does not are tied over a whole interval of priors, so indifference is not rare. Fix: state the stronger assumption, which the web appendix already does.

3 Why it matters

Bayesian Persuasion is among the most cited papers in modern microeconomic theory, and its results are used in teaching and in hundreds of applications. None of these errors affects its main theorems. Two are worth knowing in practice. The "concave and concave" rule of thumb is the kind of shortcut applied papers and lecture notes reuse, and it gives the wrong answer whenever the sender dislikes higher actions. The web-appendix formula is the reference result for persuasion with continuous states, and it is now cited in that form.

Notes

We found no published correction for these points; we have not searched exhaustively, and specialists may know some of them. All counterexamples were checked analytically and with exact rational arithmetic. Erratum 1 concerns the October 2010 web appendix; the others concern the published article.

References. Dworczak, P. and A. Kolotilin, The persuasion duality, working paper. Kamenica, E. and M. Gentzkow (2011), Bayesian persuasion, *American Economic Review* 101(6), 2590–2615, and Web Appendix (October 2010).